

Q:8 Histogram matching, transformation function

Explain histogram matching. The intensity values of an image have the PDF as:

Equation of $p(r)$

Find the transformation function that will produce an image whose intensity PDF is

Equation of $p(z)$

Histogram Matching

Histogram Matching is a digital image processing technique used to transform the pixel intensity distribution of an image so that it matches a specific **target (reference) histogram**. Essentially, it forces one image to adopt the **tone and contrast** of another.

Key Applications:

Image Normalization: It is used to standardize images taken under different lighting or sensor conditions, making them consistent for analysis.

Feature Matching: In computer vision, it helps algorithms compare images more accurately by removing variations in appearance caused by environment or exposure.

Contrast Adjustment: It allows an image to adopt the aesthetic "look" or brightness distribution of a high-quality reference image.

Transformation function

Solving for the Transformation Function

Given the intensity PDFs provided in the example for continuous intensity values:

1. Input Image PDF ($p_r(r)$):

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2} & \text{for } 0 \leq r \leq (L-1) \\ 0 & \text{otherwise} \end{cases}$$



2. Target Image PDF ($p_z(z)$):

$$p_z(z) = \begin{cases} \frac{3z^2}{(L-1)^3} & \text{for } 0 \leq z \leq (L-1) \\ 0 & \text{otherwise} \end{cases}$$



Step 1: Find the Histogram Equalization for the Input

Find the transformation s by integrating the input PDF:

$$s = T(r) = \int_0^r p_r(w) dw = \int_0^r \frac{2w}{(L-1)^2} dw = \frac{r^2}{(L-1)^2}$$



Step 2: Find the Histogram Equalization for the Target

Find the transformation $G(z)$ by integrating the specified PDF:

$$G(z) = \int_0^z p_z(w) dw = \int_0^z \frac{3w^2}{(L-1)^3} dw = \frac{z^3}{(L-1)^3}$$

Step 3: Find the Transformation Function

To find the mapping from the source to the target, we set $G(z) = s$ and solve for z :

$$\frac{z^3}{(L-1)^3} = s$$

$$z = [s(L-1)^3]^{1/3}$$

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Substituting $s = \frac{r^2}{(L-1)^2}$ into the equation:

$$z = \left[\frac{r^2}{(L-1)^2} (L-1)^3 \right]^{1/3}$$

$$z = [r^2(L-1)]^{1/3}$$

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